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Coarse woody debris decay rates for seven indigenous tree species in the central North Island of New Zealand

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ABSTRACT

The decomposition rate of stem and branch coarse woody debris (CWD, >10 cm in diameter) was assessed in natural forests located in the central North Island of New Zealand. CWD samples had originated from windfalls associated with cyclone Bernie, and had been decaying for approximately 20 years on the forest floor. Species-specific decay rates were estimated from the density of CWD samples relative to the density of live tree samples from the same stands. Decay rates were determined for rimu (*Dacrydium cupressinum*), matai (*Prumnopitys taxifolia*), tawa (*Beilschmiedia tawa*), miro (*Prumnopitys ferruginea*) and kahikatea (*Dacrycarpus dacrydioides*) in podocarp forest at Whirinaki, and red (*Nothofagus fusca*) and silver beech (*Nothofagus menziesii*) in beech forest at Kaimanawa. The average decay rate for these seven species, expressed as the time taken to lose 50% mass ($t_{1/2}$), was 30 years. Larger trees (90 cm diameter at breast height, dbh) decayed more slowly ($t_{1/2}$ = 38 years) than smaller trees (30 cm dbh; $t_{1/2}$ = 21 years). After adjustment for dbh, *P. taxifolia* ($t_{1/2}$ = 39 years), *N. fusca* ($t_{1/2}$ = 38 years), *N. menziesii* ($t_{1/2}$ = 31 years) and *B. tawa* ($t_{1/2}$ = 26 years) decayed significantly more slowly than *D. cupressinum* ($t_{1/2}$ = 18 years). *D. cupressinum* decayed more slowly than *P. ferruginea* ($t_{1/2}$ = 16 years) and *D. dacrydioides* ($t_{1/2}$ = 14 years), although these differences were not statistically significant because the CWD sample size was small for the latter two species. An attempt to expand the range of species studied using data from in-ground durability tests was not successful as species decay rankings from these tests were inconsistent with natural forest CWD rankings. Stems heavily colonized by the common decay fungus *Ganoderma* cf. *applanatum* decayed more rapidly ($t_{1/2}$ = 20 years) than those which were occupied only by other decay fungi ($t_{1/2}$ = 40 years). A tree species and dbh-dependent decay constant, λ, was derived for estimating carbon loss from CWD due to fungal decay and insect activity in indigenous forests. Future research will aim to improve these decay equations by investigating the decomposition processes associated with tree species and basidiomycete populations present at other sites in New Zealand.

1. Introduction

Coarse woody debris (CWD) can comprise a significant proportion of the carbon stock in mature forest ecosystems in New Zealand and elsewhere (Beets, 1980; Levett et al., 1985; Benecke and Nordmeyer, 1982; Schonenberger, 1984; Harmon et al., 1995; Allen et al., 1997; Eaton and Lawrence, 2006). Large quantities of woody debris are deposited periodically on the forest floor during severe storms (see for example Keller et al., 2004; Ryall et al., 2006; Woodall and Nagel, 2007). The effect of severe storms on forests was noted in an investigation initiated following cyclone Bernie, in April 1982, when it was concluded that they have had a major influence on stand structure and pattern in indigenous forests throughout New Zealand (Shaw, 1983). Carbon losses due to CWD decay are therefore an important aspect of the terrestrial carbon balance.

New Zealand has determined under the 1992 United Nations Framework Convention on Climate Change to quantify the levels of carbon exchange occurring within its portion of the terrestrial biosphere, and research has commenced to quantify the organic carbon balance in indigenous forest and shrubland, including changes over time. Initial research to establish the 1990 baseline (Hall et al., 2001) has identified a number of sampling and carbon estimation issues. These are being addressed by implementing a national inventory, which involves mapping the area of indigenous forest and shrubland, establishing a network of permanent vegetation plots (Coomes et al., 2002) on an 8 km grid placed over the mapped area, measuring the dimensions of the trees,

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shrubs, and coarse woody debris in all plots over a five year period following standard protocols (Payton et al., 2004), and concurrently measuring fine litter, humus and mineral soil carbon to 30 cm depth on about one third of the plots (Davis et al., 2004).

In accordance with IPCC Good Practice Guidance for Land Use, Land-Use Change and Forestry (2003), carbon stocks in New Zealand's natural forest and shrubland are being estimated in five pools: (1) above ground biomass, (2) below ground biomass, (3) dead wood, (4) litter, and (5) mineral soil. A national estimate of above ground biomass carbon in New Zealand's natural forest estate is reported in Hall et al. (2001) and total carbon within forest and shrubland in a 60 km wide transect across the South Island of New Zealand are reported in Coomes et al. (2002). A few site-specific studies in New Zealand have provided data suitable for developing regression relationships for predicting biomass from tree and log dimensions (Beets, 1980) and it is acknowledged that the improvement of these functions is highly desirable.

Estimates of CWD carbon stocks are currently based on measurements of the debris volume over bark, which are calculated from lengths and small and large end diameters, and an assumed density, with modifiers used to account for reductions in CWD density in subjectively assessed decay classes (Coomes et al., 2002; Harmon et al., 1995; Waddell, 2002; Eaton and Lawrence, 2006). Carbon is then estimated from the CWD mass using a conversion factor of 0.50 (Allen et al., 1997). Changes in CWD carbon stocks over time can therefore be estimated from repeated measurements of CWD volume by decay class. Remeasurement of residual debris volume allows for fragmentation and decay of existing CWD and for the addition of new CWD due to tree mortality (Stone et al., 1998). A measurement based approach for estimating CWD carbon stock changes over time entails ongoing remeasurement costs and can be inaccurate, for example, following major windthrow events considerable quantities of material are added to the forest floor and can be difficult to locate and remeasure consistently over time. Furthermore, CWD decay classes are a subjective assessment of external features, and may not accurately depict CWD density (Creed et al., 2004).

An alternative modelling approach for estimating CWD stocks and changes in plots that are remeasured over time involves documenting when tree mortality events occur, estimating the initial volume, whole stem density, and resultant mass of fallen trees using appropriate biomass equations developed for live trees, and then applying decay functions to estimate the loss in mass over time. Studies undertaken elsewhere have shown that decay rates differ among tree species, and that factors including the diameter of the debris, and climate (temperature) are important considerations when developing decay functions, for example, Frangi et al. (1997), Chambers et al. (2000), Mackensen et al. (2003), Busing and Fujimori (2005), Wilson and McComb (2005), and Eaton and Lawrence (2006). Decay functions therefore need to be applied to plots taking into account the tree species, diameter at breast height, CWD age, and mean annual temperature.

CWD decay rates for New Zealand tree species have not been determined under forest conditions. A large number of species are involved, and our intention was to measure decay rates for species that comprise a large proportion of the carbon stock in New Zealand's natural forest, and estimate decay rates for less common species based on their expected natural durability ratings.

The durability of heartwood of a wide range of New Zealand's timber trees has been determined using a universal system of testing involving 50 mm × 50 mm heartwood stakes placed in the ground in test areas referred to as graveyards (Clifton, 1994), following a classification scheme used routinely in Australia (Page et al., 1997). Based on this system, New Zealand indigenous tree species have been broadly classified into five durability classes ranging from perishable (<5 years) to very durable (>25 years), which may provide an initial basis for grouping species and allocating decay constants to these classes, as attempted in Australia (Mackensen et al., 2003). It is currently not known how closely species durability ratings based on graveyard studies compare with decay rates obtained under forest conditions.

Variation in CWD decay rates owing to the particular fungal species involved has been reported previously, although few studies have been undertaken on this topic (Worrall et al., 1997; Edman et al., 2006; Buchanan et al., 2001). As part of this study we therefore quantified the influence of basidiomycete fungal species populating the CWD on decay rates, following assessment methods published previously (Hood et al., 2004).

Research has shown that accurate estimation of CWD decay rates requires the use of methods that take account of losses owing to fragmentation of stems and branches while allowing for any decay that occurred in damaged trees prior to tree mortality (Means et al., 1985; Harmon and Sexton, 1996; Stone et al., 1998; Creed et al., 2004). Practical guidelines for measuring CWD, and for determining unbiased decay rates using chronosequence (space for time substitution) and age series approaches are given in Harmon and Sexton (1996). Decay rates can be estimated accurately provided that wood density is measured consistently by taking into account voids associated with stem fluting and hollowing (Palace et al., 2007) using mensurational methods that apply both to live trees and CWD. The volume of samples immediately prior to mortality should be used to calculate the density of CWD. Furthermore, the precision of the decay constant estimates can be improved by allowing for internal log decay that may have occurred prior to tree mortality. The methods described in this paper address these issues, and also incorporate double sampling which facilitates the development of more precise decay constant estimates than otherwise possible when the sample size is relatively small, as in this study.

The objectives of this paper are to provide new CWD decay data and models for estimating species-specific decay rates for major indigenous tree species growing in the North Island of New Zealand, and to identify the main fungal agencies involved, in order to facilitate national predictions of carbon loss following tree mortality using carbon models.

2. Materials and methods

2.1. Study sites

Sites where cyclone Bernie resulted in substantial areas of tree mortality in 1982 were identified in Whirinaki and Kaimanawa forests, each dominated by different tree species.

2.1.1. Whirinaki forest

This study site included a large trial (40 ha) established in 1979 in a dense podocarp forest in Whirinaki forest near Minginui (latitude 38°39′S, longitude 176°42′E) at an altitude of 460 m (Smale et al., 1985, 1998). Precipitation at Whirinaki averages ca. 1500 mm per year, with a comparatively large annual variation, while the mean annual temperature (MAT) is 11 °C. The terrain is flat to gently rolling, and the soil parent material is derived from ejecta from the Taupo volcanic centre. The present forest has arisen since the last cataclysmic Taupo eruption approximately 1800 years BP. The height of live trees measured for the wood density survey undertaken as part of the study, indicated that the canopy ranges between 45 and 60 m in height, and is dominated by rimu (*Dacrydium cupressinum* Lamb.) and matai (*Prumnopitys taxifolia* (D. Don) de Laub.), with occasional kahikatea (*Dacrycarpus dacrydioides* (A. Rich.) de Laub.) and totara (*Podocarpus totara* D.

Don). These podocarps emerge above a subcanopy of miro (*Prumnopitys ferruginea* (D. Don) de Laub.), tawa (*Beilschmiedia tawa* (A. Cunn.) Kirk) and other minor hardwood species up to 35 m in height.

2.1.2. Kaimanawa forest

This study site is situated in beech forest in Kaimanawa forest (latitude 38°57′S, longitude 176°13′E) at an altitude of approximately 750 m. Precipitation averages ca 1700 mm per year, while the MAT is 9.5 °C. The terrain at this site is undulating, and as at Whirinaki, the soil parent material is derived from ejecta from the Taupo volcanic centre, the last major episode being the Taupo eruption (ca. 1800 years BP). The present forest canopy ranges between 25 and 35 m in height, and is dominated by red beech (*Nothofagus fusca* (Hook. f.) Oerst) and silver beech (*Nothofagus menziesii* (Hook. f.) Oerst.). Gaps created by previous windthrow events are occupied predominantly by even-aged pole beech stands. Two study locations between 700 and 800 m altitude impacted by cyclone Bernie were identified to cover a range in tree size, including a pole stand near the old mill site with small diameter fallen trees and an area of mature forest nearby with large diameter fallen trees of both species.

2.2. CWD sampling

The tree species selected for study included *D. cupressinum*, *P. ferruginea*, *P. taxifolia*, *D. dacrydioides*, *B. tawa*, *N. fusca*, and *N. menziesii*. These species were classified as perishable (*B. tawa*), non-durable (*P. taxifolia*, *P. ferruginea*), moderately durable (*N. menziesii*, *D. cupressinum*), and durable (*N. fusca*) based on an international durability rating system (Clifton, 1994). CWD samples were taken from trees uprooted mostly in a large storm in 1982, although a few trees were included that fell in later storms. The samples from Whirinaki included *D. cupressinum* (eight trees) and *P. taxifolia* (eight trees) that were uprooted in 1982 and for which stem diameters had previously been measured at intervals along the stem and disc samples assessed for decay fungi in 1986 by Hood et al. (1989). These 16 trees were re-sampled to assess decay fungi and wood density in 2001 (2 trees) and 2002 (14 trees, reported in Hood et al., 2004). Additional tree species assessed included *B. tawa* (five trees blown down in 1983 and one tree in 1997), *P. ferruginea* (one tree in 1983), and *D. dacrydioides* (one tree in 1982), which were all sampled in 2007. Stems lay in heavy shade, the majority in ground contact throughout the full period from the time of windfall. CWD samples from Kaimanawa included *N. fusca* (14 stems) and *N. menziesii* (9 stems) that were blown down during the 1982 storm and were sampled in 2005. Samples for the assessment of decay fungi were taken from 12 of these stems (6 of each species) early in 2006.

2.3. CWD measurements

For each fallen tree, stem diameter over bark was measured at 0.15, 0.7, 1.4 m (diameter at breast height, dbh), 3.0 m, and thereafter at 3.0 m intervals along the stem, including forks and branches up to a 10-cm diameter top. When measuring stem and branch diameters, care was taken to determine the original diameter, by making allowance for any bark and sapwood loss that had occurred due to fragmentation. Bark and sapwood measurements from sound parts of the same log and measurements made on recent windfalls were used as a guide when estimating fragmentation losses. Stem diameter measurements of 16 trees made 4 years after windthrow by Hood et al. (1989) provided 47 measurements that could be compared with ours. The mean difference between the two measurement periods was only 4 mm (S.E. = 6 mm), which was not significantly different from zero ($t_{47} = 0.69$, $p = 0.49$), and provided confidence in the approach we followed. Height to the 10 cm stem diameter cut-off point was then measured. Stem disc samples were cut at 1.4, 6.0 m and thereafter at 6.0 m intervals along the stem, including from major forks.

To provide data suitable for estimating decay rates, CWD density was determined following mensurational methods developed for live trees (Beets and Pollock, 1987; Creed et al., 2004), with some modifications incorporated to allow for internal log decay where this evidently occurred prior to tree mortality. Discs were cut uniformly ca. 40 mm thick, ensuring that all loose bark and associated wood pieces were retained. Prior to oven drying, disc thickness was measured with calipers, at 4, 6 or 8 equidistant marked points around the circumference depending on disc diameter. A tape was aligned from the centre of disc to each of the marked points and the radius of the hollow centre (in rare instances where void occurred) and heartwood was measured. If the hollow part of the disc was irregular in shape, up to 10 additional radial measurements were made. If hollowing resulted from decay prior to tree mortality, the hollow area was deducted from the disc total original cross-sectional area, when calculating disc volume. Lengths and the original large and small end diameters of all branches >10 cm diameter were measured as for stems. Discs ca. 40 mm thick were cut from branches at either 3 or 6 m intervals, depending on branch cumulative length, with a maximum of 10 branch discs taken per tree. Branch discs were measured and processed as for stem discs. Total dry weight of discs was measured after drying in forced ventilation ovens at 70 °C to constant weight. The density (wood plus bark combined) of stem and branch (>10 cm diameter) disc samples was calculated by dividing the disc oven dry weight by the disc original volume. The displacement volume of presoaked or sealed discs has sometimes been used by other researchers to estimate disc density, however the mensurational approach avoids bias associated with the displacement method when stems are fluted or the bark fissured (Mackensen et al., 2003; Creed et al., 2004; Beets et al., 2007). No data were collected from stems or branches <10 cm diameter, which was defined as fine woody debris (Coomes et al., 2002).

2.4. Fungal agencies involved in decay

To assess decay fungi, additional discs were cut from the CWD adjacent to a proportion of those measured for density. Sub-samples were taken from either four (2001, 2002, and 2006) or two (2007) radial sectors around the circumference of each disc. Isolations were attempted in a laminar flow cabinet from small (ca. 1 mm diameter) wood pieces cut at measured points along the sector length after first splitting the sector to expose an uncontaminated surface. Wood pieces were plated onto an isolation medium, and emerging fungi were cultured as described in Hood et al. (2004). Decay fungi obtained in this way were identified mostly to species level for each disc within three radial depth zones, 0–6, 6–12, and >12 cm, measured from the disc perimeter (Hood et al., 2004). The total yield for each tree was calculated as the number of points from which decay fungi were isolated divided by the total number of points from which an isolation was attempted for the tree. A prevalence value for each of the major fungal species was obtained for each tree as the proportion of discs from the tree with at least one positive sample for the species.

2.5. Sampling of recently fallen trees

Determination of the decay rate of CWD required an estimate of the initial density of the stems and branches at the time of tree fall.

Because tree felling was not possible, the initial density was estimated using a double sampling procedure (Cochran, 1977), where the ratio of whole stem and branch (>10 cm) wood and bark density to outer wood basic density of recently fallen trees was used to estimate the initial density of CWD based on a breast height outer wood density survey of live standing trees. The whole stem and branch (>10 cm) wood and bark density of recently fallen trees was measured using identical procedures as described for the CWD. Discs were cut at 1.4 m height and thereafter at 6 m intervals along the length of the stem and branches (>10 cm diameter). Trees were considered as recently fallen if they had green foliage still attached (assumed to have fallen in the current year), or if frass from current insect activity and foliage colour and appearance suggested that wind fall occurred between 1 and 5 years ago. In addition, recently fallen trees were allocated to one of three decay classes, based on the condition of the sapwood, with 1 indicating that decay was absent in all discs, 2 that sapwood was sound but fungal attack was evident in some discs, or 3 that sapwood had collapsed (soft/breaks) owing to fungal decay, which was mostly confined to upper stem and branch discs. The mean wood and bark density of each tree was estimated as the area-weighted average density of the discs, excluding hollowing associated with internal decay that occurred prior to tree mortality. Breast height outer wood basic density was measured at 0–5 and 5–15 cm depths under bark in block samples cut from each recently fallen tree. Basic wood density is defined as the oven dry weight of a sample divided by its green (saturated) volume (Clifton, 1994). The fresh volume of the block sample was measured by water displacement and the sample was oven dried to constant weight at 105 °C. Double sampling aimed to improve the precision of decay constants developed for these seven tree species, using a large sample of trees with breast height outer wood density measurements.

2.6. Outerwood basic density survey

The site mean outer wood basic density estimate was based on a breast height core taken from a sample of 20 live standing trees per species growing in the vicinity of the recently fallen trees. As a contingency, to allow for within species variation in density with tree size, total height of each tree was also measured with a hypsometer and dbh was measured with a diameter tape. Outer wood density was defined as the basic density of wood from 5 to 15 cm depth, measured under the bark, and was determined using the maximum moisture content method (Smith, 1954). The outer 5 cm of sapwood was not used as a ratio estimator because this was usually affected by pinhole borer in the recently fallen trees, while heartwood is not attacked (Clifton, 1994).

2.7. Graveyard study

The durability of heartwood has been examined in a wide range of species growing in New Zealand, following a testing procedure using 50 mm × 50 mm heartwood stakes partially inserted into the soil in test areas referred to as graveyards (Clifton, 1994). The correlation between durability ratings from the graveyard study, estimated as described in the following section, and decay constants developed under forest conditions, was examined to determine whether the graveyard study could provide a basis for estimating decay constants for species with no decay data. Graveyard studies document the number of years taken for each heartwood stake to break. This follows a standard test procedure where annually each stake is knocked sharply on two adjacent sides with a wooden mallet (Clifton, 1994). Graveyard studies were installed in 1951, 1955, 1967, and 1970 at six sites throughout New Zealand covering a wide range of climate and soil types, and including a wide range of species, although only a subset of species were well represented at each site. A proportion of the stakes from the more durable species had not broken when the study sites were last assessed.

2.8. Statistical analysis

2.8.1. Live tree wood density

The ratio of whole stem and branch (>10 cm) wood and bark density to outer wood basic density of recently fallen trees was used to estimate the whole stem and branch (wood and bark) density of survey trees for each species. These ratio estimates $\bar{D}_R$ were calculated using the following equation (e.g., Cochran, 1977):

$$\bar{D}_R = R\bar{D}_O \tag{1}$$

where $R = \bar{d}_T/\bar{d}_O$ is the ratio of the mean stem and branch (>10 cm) wood and bark density ($\bar{d}_T$) to the mean outer wood basic density ($\bar{d}_O$) of the recently fallen trees, and $\bar{D}_O$ is the mean outer wood basic density of the surveyed live trees. Standard errors of these estimates were calculated using (Cochran, 1977):

$$\text{S.E.}(\bar{D}_R) = \sqrt{\frac{S_T^2 - 2RS_{TO} + R^2S_O^2}{n} + \frac{R^2S_O^2}{N}} \tag{2}$$

where S_T^2, S_O^2, and S_{TO} are the variances of wood and bark density, outer wood basic density and their covariance from the recently fallen trees, n is the number of recently fallen trees of all species ($n = 29$), and N is the number of surveyed trees ($N = 20$ for each species; $N = 140$ across all species). To allow for a possible loss in density due to decay evident in some of the recently wind blow trees, the effect of number of years from tree fall and sampling, and also of decay class on R was examined.

2.8.2. Decay rates

Mean whole stem and branch wood and bark (>10 cm) density of CWD trees ($\bar{D}_{CWD}$) and their standard errors were calculated for each species using the SAS (Windows Version 9, SAS Institute Inc., Cary, NC, USA) procedure GLM. Species means were compared using the least significant difference (LSD) test. Decay rates were estimated from the measured mean whole stem and branch density of CWD and the estimated mean whole stem and branch density of the survey trees. An exponential decay model was assumed, i.e., it was presupposed that density at time t years after falling is given by

$$\bar{D}_{CWD} = \bar{D}_R \exp(-\lambda t) \tag{3}$$

where, in this case, $\bar{D}_R$ is the density of live trees estimated using Eq. (1), and λ is the decay constant (Olson, 1963). Decay constants were estimated for each species and across all species using the following equation:

$$\lambda = \frac{\ln(\bar{D}_R) - \ln(\bar{D}_{CWD})}{\bar{t}} \tag{4}$$

where $\bar{t}$ is the mean time since falling and sampling of the CWD trees. The overall mean decay rate across species will be applied to those without data, unless species durability ranks in the graveyard study are reasonably strongly correlated with forest decay rates. Standard errors of these decay constants were estimated by a formula derived using the delta method (Oehlert, 1992):

$$\text{S.E.}(\lambda) = \frac{1}{\bar{t}}\sqrt{\left(\frac{\text{S.E.}(\bar{D}_R)}{\bar{D}_R}\right)^2 + \left(\frac{\text{S.E.}(\bar{D}_{CWD})}{\bar{D}_{CWD}}\right)^2} \tag{5}$$

The time taken to lose 50% density was estimated from the decay constant:

$$t_{1/2} = \frac{\ln(2)}{\lambda} \quad (6)$$

with standard error obtained using the delta method:

$$\text{S.E.}(t_{1/2}) = \frac{\text{S.E.}(\lambda)\ln(2)}{\lambda^2} \quad (7)$$

2.8.3. Dbh-dependent decay rates

There was a tendency for density of both live and CWD trees to be related to dbh, and because these two categories of tree differed in mean dbh it was desirable to also incorporate tree size effects into the decay constant equations. The following method was used to achieve this. Firstly, power functions of the following form were fitted to both outer wood basic density of live trees, and whole stem and branch density of CWD trees:

$$D_{ij} = D'_i\left(\frac{\text{dbh}_{ij}}{\text{dbh}_{\text{std}}}\right)^b \quad (8)$$

where dbh_{std} is a standardized value of dbh (61.9 cm, the mean dbh of the CWD trees was chosen), D_{ij} and dbh_{ij} are the density and dbh of the jth tree of the ith species, the parameter b quantifies the effect of dbh on density (e.g., a zero value of b signifies no effect, while a positive or negative relationship is represented by a positive or negative value for b), and, the parameters D'_i are the whole stem and branch densities for each species adjusted to the standardized dbh. This model was fitted using the SAS procedure NLIN with the parameters D'_i being estimated using dummy variables. Model (8) was fitted for both outer wood basic density of live survey trees, and density of CWD trees, producing estimates for b of b_{O} and b_{CWD}, respectively, and dbh-adjusted outer wood and CWD whole stem and branch (>10 cm) densities for each species of D'_{O} and D'_{CWD}. The D'_{O} values were then converted into stem and bark densities for live trees using ratio estimates (Eq. (1)), and these were combined with D'_{CWD} in Eq. (4) to produce decay constants adjusted to the standardized dbh for each species and across all species:

$$\lambda' = \frac{\ln(RD'_{\text{O}}) - \ln(D'_{\text{CWD}})}{\bar{t}} \quad (9)$$

These decay constants adjusted to a standardized dbh were then used to derive dbh-dependent decay constants, $\lambda(\text{dbh})$, where:

$$\lambda(\text{dbh}) = \frac{\lambda' + (b_{\text{O}} - b_{\text{CWD}})}{\bar{t}}\ln\left(\frac{\text{dbh}}{\text{dbh}_{\text{std}}}\right) \quad (10)$$

2.8.4. Fungal agencies involved in decay

Relationships between wood density of CWD trees and the yield and mix of fungal species present on these trees were explored using a variety of linear and nonlinear models fitted using the SAS procedures GLM and NLIN. All these models were fitted at the tree-mean level, with wood density as the dependent variable. Overall yield of decay fungi from the tree, and the prevalence value of each fungal species were included as independent variables. Also included in these models was tree host species, and tree dbh, with the latter variable testing for a relationship between decay rate and tree size. These analyses aimed to test for host tree species, tree size, and fungal species effects on wood density, and by implication, decay rate.

2.8.5. National graveyard study

Clifton (1994) assigned species to durability classes, which were considered too broad for our purposes. The most recent stake data were acquired and the effect of tree species and site on the mean time in years taken for stakes to break was analysed using the SAS procedure LIFEREG, using time to failure as the dependent variable. This procedure allows for the fact that for many species, not all stakes had broken at the most recent assessment date. Times taken for 50% of sample stakes to fail in graveyard trials were estimated by species, and these were related to decay rates obtained under forest conditions.

3. Results

3.1. Densities of live trees and CWD

The number of sample trees, and their mean dimensions are summarised in Table 1. Note that the predominant podocarps at Whirinaki were generally larger in height and diameter than the beech species at Kaimanawa. Tree height and diameter data for standing live trees are shown in Fig. 1. The live tree survey data showed that breast height outer wood basic density varied both between and within species. Within a species density varied by about ±50 to ±75 kg m^{-3}, with a weak trend of density reducing with increasing dbh (Fig. 2). The 20 survey trees per species provided estimates of mean outer wood basic density at breast height within less than ±3% (95% CI) with species means ranging from 389 kg m^{-3} for *D. dacryoides* to 543 kg m^{-3} for *P. taxifolia* (Table 2). To account for the decreasing trend of density with increasing dbh (Fig. 2), Eq. (8) was fitted to the outer wood density data. The parameter b_{O} was significantly less than zero ($b_{\text{O}} = -0.0795$; S.E. = 0.018; t-test $t_{132} = -4.42$, $p < 0.0001$).

Mean stem and branch (>10 cm) wood and bark densities for recently fallen trees and CWD are shown in Table 2. For recently fallen trees, the ratio of mean stem and branch (>10 cm) wood and bark density to mean breast height outer wood basic density

Table 1
Mean dimensions of survey trees, recently fallen trees, and CWD, and the mean time since the windfall event

Location	Species	Survey trees			Recently fallen trees			CWD trees			
		n	dbh (cm)	Ht (m)	n	dbh (cm)	Ht (m)	n	dbh (cm)	Ht (m)	Mean time (year)
Whirinaki	*Beilschmiedia tawa*	20	43	28	4	25	24	6	45	29	22
	Dacrydium cupressinum	20	71	26	10	102	49	8	92	51	20
	Dacrycarpus dacrydioides	20	121	49	1	142	54	1	138	50	25
	Prumnopitys taxifolia	20	82	41	4	61	37	8	83	44	20
	Prumnopitys ferruginea	20	54	34	6	51	35	1	40	32	24
Kaimanawa	*Nothofagus fusca*	20	61	27	2	25	17	14	42	25	23
	Nothofagus menziesii	20	71	26	2	87	29	9	54	27	23
Total/average across all trees		140	78	37	29	70	38	47	62	32	22

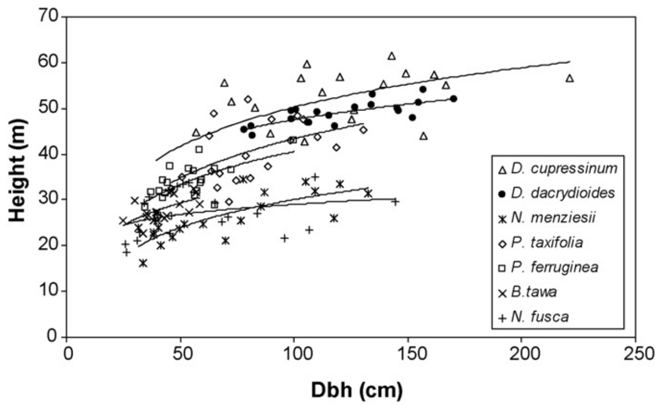

Fig. 1. Tree total height shown in relation to dbh of 20 survey trees of each species at Whirinaki and Kaimanawa forests in the Central North Island, New Zealand.

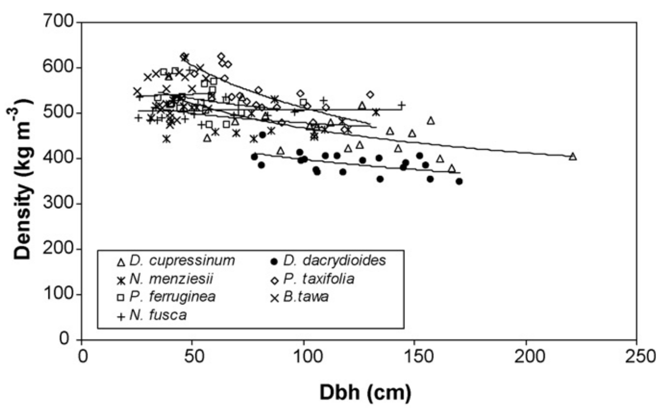

Fig. 2. Breast height outer wood basic density shown in relation to dbh of 20 survey trees of each species at Whirinaki and Kaimanawa in the Central North Island, New Zealand.

(5–15 cm depth) used in Eqs. (1) and (2) was not affected by the number of years since it had fallen (F-test $F_{1,27} = 1.40$, $p = 0.25$), nor by its stem decay class ($F_{2,26} = 0.35$, $p = 0.71$). The mean (and standard error) of the ratio, R, was 0.918 (0.018), 0.913 (0.027) and 0.895 (0.022) for decay classes 1, 2 and 3, respectively, with the ratio decreasing slightly with increasing decay class. The ratio did not differ significantly between species ($F_{6,22} = 0.58$, $p = 0.74$), and averaged 0.910 (S.E. 0.012) across all species. The correlation coefficient between breast height outer wood basic density and whole stem and branch (>10 cm) wood and bark density was 0.87 (Fig. 3). The variances and covariance required in Eq. (2) were $S_T^2 = 3178$, $S_O^2 = 4749$, and $S_{TO} = 3374$. Stem and branch (>10 cm) wood and bark densities of the recently fallen trees, adjusted by the mean breast height outer wood basic density from the live tree survey using the ratio revealed significant differences between species (based on LSD test of the ratio estimates in Table 2). The stem and branch (>10 cm) wood and bark of *P. taxifolia*, *B. tawa*, and to a lesser extent, *P. ferruginea* were significantly denser than those of the other species, while *D. cupressinum* and more so, *D. dacrydioides*, showed the lowest initial densities (Table 2). Densities of the beeches (*Nothofagus* spp.) were intermediate. The survey adjusted densities are regarded as the best currently available estimates of the mean initial whole stem and branch (>10 cm) wood and bark density for these species at each location.

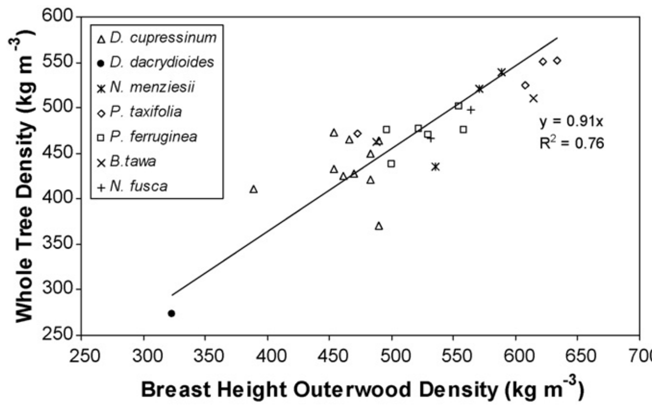

Fig. 3. Whole stem and branch (>10 cm) wood and bark density shown in relation to breast height outer wood basic density for recently fallen trees, based on disc samples obtained soon after windthrow.

Whole stem and branch (>10 cm) densities of CWD were substantially lower for all species approximately 20 years after windthrow (Table 2). Individual disc densities from *P. taxifolia* and *D. cupressinum* CWD and recently fallen trees varied both within and between trees, and were lower for CWD (Fig. 4(a) and (b)). After 20 years on the forest floor, *D. cupressinum* and *P. taxifolia* disc densities averaged 57% and 75%, respectively, of the adjusted mean initial densities, with some dependence on height up the tree (Fig. 5). These reductions in density were associated with a reduction in disc dry weights due to fungal decay and insect activity, since CWD volume was based on the over-bark disc diameter.

Table 2
Breast height outer wood basic density of survey trees, and whole stem and branch (>10 cm) wood and bark density (kg m^{-3}) of recently fallen trees, and of CWD (sample size n, Table 1) including a more precise ratio estimate of the initial whole stem and branch wood and bark density of recently fallen trees after adjustment based on the outer wood basic density survey

Location	Species	Breast height outer wood basic density survey trees		Stem and branch density, recently fallen trees				Stem and branch density, CWD trees	
		Mean	S.E.	Mean	S.E.	Ratio estimate	S.E.	Mean	S.E.
Whirinaki	*B. tawa*	541a	9	495ab	16	490a	10	250bc	21
	D. cupressinum	461c	9	434c	10	418d	10	226bcd	18
	D. dacrydioides	389d	9	273d	32	353e	10	160cd	50
	P. taxifolia	543a	9	525a	16	492a	10	367a	18
	P. ferruginea	531a	9	473bc	13	481ab	10	138d	50
Kaimanawa	*N. fusca*	506b	9	482abc	23	459bc	10	267b	13
	N. menziesii	492b	9	487ab	23	446c	10	255bc	17
Average across all trees		495	3	464	6	448	6	270	7

Values in a column followed by the same letter do not differ significantly (LSD test; $\alpha = 0.05$).

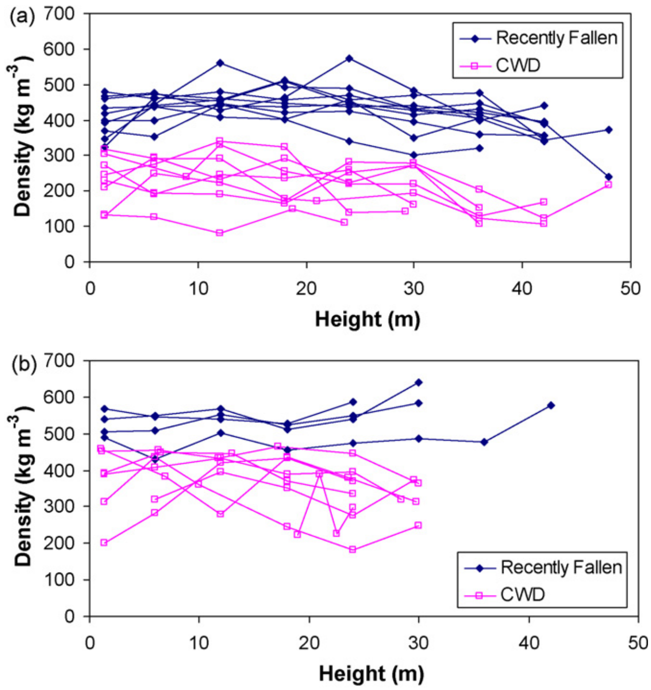

Fig. 4. Density of disc samples from recently fallen trees and CWD, with positions along stem indicated as height above original ground level in the standing tree. Discs for CWD study were cut from trees uprooted in 1982: (a) *Dacrydium cupressinum*; (b) *Prumnopitys taxifolia*.

As noted above, smaller diameter live trees tended to have higher outerwood densities than larger diameter trees of the same species (Fig. 2). However, the reverse was true for CWD whole tree densities with larger diameter trees tending to be more dense than smaller trees. The effect was statistically highly significant when the mean diameter of discs in each stem was used as the independent variable in Eq. (8), but slightly less so when dbh was used ($b_{CWD} = 0.221$; S.E. = 0.068; $t_{39} = 3.23$, $p = 0.0025$).

3.2. Decay rates

Decay constants λ, unadjusted for dbh are given in Table 3 and the corresponding times taken for CWD to lose 50% of initial density are given in Fig. 6, by species. The decay constant for *P. taxifolia* was significantly lower than the constant for *B. tawa*, *D. cupressinum*, *D. dacrydioides*, *N. fusca*, *N. menziesii* and *P. ferruginea*,

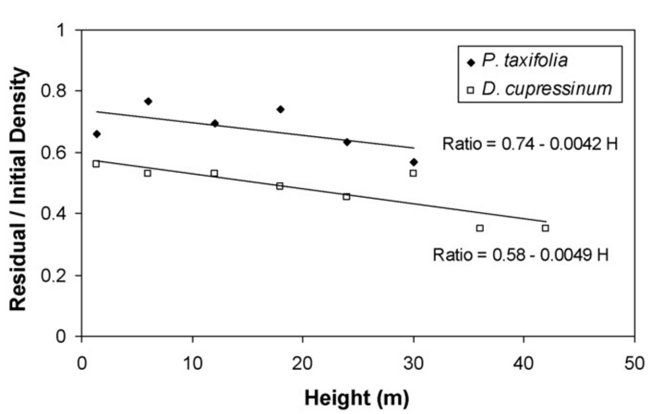

Fig. 5. Ratio of means of disc density in CWD/recently fallen trees given in relation to tree height. Discs for estimating residual density were cut from CWD trees uprooted in 1982.

Table 3
Decay constants λ and time $t_{1/2}$ taken to lose 50% density by species

Location	Species	λ	S.E.	$t_{1/2}$ (year)	S.E.
Whirinaki	*B. tawa*	0.0312 b	0.0039	22.2	2.8
	D. cupressinum	0.0308 b	0.0041	22.5	3.0
	D. dacrydioides	0.0316 ab	0.0127	21.9	8.8
	P. taxifolia	0.0147 a	0.0026	47.1	8.2
	P. ferruginea	0.0520 b	0.0153	13.3	3.9
Kaimanawa	*N. fusca*	0.0235 b	0.0023	29.5	2.9
	N. menziesii	0.0243 b	0.0030	28.5	3.5
Average across all trees		0.0233	0.0013	29.8	1.6

Values in a column followed by the same letter do not differ significantly (LSD test; $\alpha = 0.05$).

with no other significant differences found among species. The decay constant averaged across all species and its standard error was $\lambda = 0.0233 \pm 0.0013$, which implies that it takes on average 29.8 ± 1.6 years for a fallen tree to lose half its mass.

3.3. Dbh-dependant decay rates

The contrasting negative and positive relationships between wood density and dbh for respectively the live and CWD trees, implies that fallen smaller diameter trees must decay more rapidly than larger trees. Decay constants for each species adjusted to a standardized dbh of 61.9 cm are shown in Table 4 and the corresponding times taken to lose 50% of initial density are shown in Fig. 6. Species rankings of dbh-adjusted and unadjusted decay constants were similar, however significant species differences were more clearly evident after adjusting for dbh. *P. taxifolia*, *N. fusca*, and *N. menziesii* decayed more slowly than *D. cupressinum*, *P. ferruginea*, and *D. dacrydioides*, while *B. tawa* had an intermediate decay rate (Table 4).

When the dbh-adjusted densities predicted using Eq. (8) for live and CWD trees were substituted into Eqs. (9) and (10), the following dbh-dependent decay constant was obtained:

$$\lambda(\mathrm{dbh}) = \lambda' + b\ln\left(\frac{\mathrm{dbh}}{61.9}\right) \qquad (11)$$

where $b = (b_O - b_{CWD})/\bar{t} = -0.0138$ (S.E. 0.0032), and λ' are decay constants adjusted to the standardized dbh of 61.9 cm. If the decay-rate differences between species observed in this study are ignored, a common adjusted decay constant of $\lambda' = 0.0235$ (Table 4) would be used in Eq. (11) for species without data. The adjusted decay constants tabulated for each species in Table 4 would be used in Eq. (11) to provide species-specific dbh-dependent decay constants for the seven species with decay data.

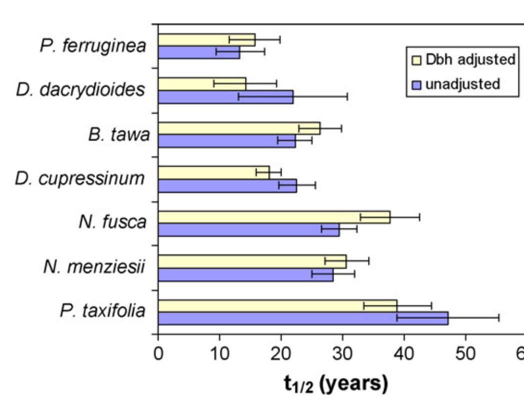

Fig. 6. Time taken for CWD to lose 50% of its original density by species, with and without adjustment for dbh. Error bars show standard errors.

Table 4
Decay constants λ and estimated time $t_{1/2}$ taken to lose 50% density by species corrected to a standardised dbh of 61.9 cm

Location	Species	λ	S.E.	$t_{1/2}$ (year)	S.E.
Whirinaki	*B. tawa*	0.0262b	0.0035	26.4	3.5
	D. cupressinum	0.0383a	0.0044	18.1	2.1
	D. dacrydioides	0.0487ab	0.0173	14.2	5.1
	P. taxifolia	0.0177b	0.0025	39.2	5.6
	P. ferruginea	0.0440ab	0.0117	15.8	4.2
Kaimanawa	*N. fusca*	0.0182b	0.0024	38.0	5.0
	N. menziesii	0.0225b	0.0026	30.8	3.6
Average across all trees		0.0235	0.0012	29.5	1.5

Values in a column followed by the same letter do not differ significantly (LSD test; α = 0.05).

3.4. Agencies involved in decay

A range of basidiomycete decay fungi were cultured from CWD at both sites, the most prevalent being *Ganoderma applanatum* sensu Wakef. (referred to here as *G.* cf. *applanatum*), *Armillaria* spp. (*A. novae-zelandiae* (G. Stevenson) Herink and *A. limonea* (G. Stevenson) Boesewinkel) and at the Kaimanawa beech forest site only, *Cyclomyces tabacinus* (Mont.) Pat. (Hood et al., 2004, 2008). For the CWD trees, there was no significant relationship between yield of all decay fungi and wood density ($t_{27} = 0.62$, $p = 0.44$). However, the fungal species present on a decaying tree were significantly implicated in the rate of decay. The prevalence value of *G.* cf. *applanatum* (the proportion of discs from which this fungus was isolated) was significantly negatively related to wood density ($t_{27} = 6.37$, $p = 0.018$). Prevalence values for the other major wood decay species in the study (*Armillaria* spp. and *C. tabacinus*) were not significantly associated with wood density. This result was consistent across tree species with no significant interaction between tree species and prevalence of *G.* cf. *applanatum*. A nonlinear regression model was fitted to predict CWD wood density as a function of dbh and prevalence of *G. applanatum*, and this was combined with the dbh-dependent live wood density model to produce the following decay constant model:

$$\lambda(\mathrm{dbh}, G) = 0.0235 - 0.0138 \ln\left(\frac{\mathrm{dbh}}{61.9}\right) + 0.0171 \times (G - 0.353) \tag{12}$$

where *G* is the prevalence of *G.* cf. *applanatum*. In this equation, the mean value of *G* (0.353) has been subtracted from the *G* term, and dbh is divided by its mean (61.9 cm), so that the value of the first parameter in the model (0.0233 from Table 3) is the mean decay constant for the CWD trees in the study. This model was used to derive decay constants for a range of tree diameters and *G.* cf. *applanatum* prevalence values (Table 5). Decay rates approximately halve, and time taken to decay approximately doubles as tree dbh increases from 30 to 90 cm. Likewise, CWD decay rates increase in the presence of *G.* cf. *applanatum* and are approximately double in trees where all discs are affected by this decay fungus.

Table 5
Effects of tree diameter and *Ganoderma applanatum* prevalence on wood density after approximately 20 years on the forest floor, decay constant λ and estimated time $t_{1/2}$ taken to lose 50% density

Factor	Level	λ	S.E.	$t_{1/2}$ (year)	S.E.
Dbh (cm)	30	0.0335	0.0027	20.7	1.6
	60	0.0239	0.0013	29.0	1.5
	90	0.0183	0.0018	37.8	3.4
G. cf. *applanatum* prevalence	0	0.0175	0.0022	39.7	4.6
(proportion of discs)	0.5	0.0260	0.0016	26.6	1.6
	1	0.0346	0.0039	20.0	2.2

Table 6
Estimated time $t_{1/2}$ taken for 50% of sample stakes to fail in graveyard trials for indigenous species

Species	$t_{1/2}$ (year)	S.E.
Nothofagus truncata (Col.) Cock.	55.9a	8.0
Podocarpus totara	39.4ab	3.2
N. fusca	33.5bc	2.7
Manoao colensoi (Hook.) Molloy	29.0cd	2.7
Nothofagus solandri var. *cliffortioides* (Hook. f.) Poole	23.0d	1.6
Elaeocarpus dentatus (J. R. et G. Forst.) Vahl	14.5e	1.0
Metrosideros spp.	14.4ef	2.3
N. menziesii	14.4ef	1.0
Libocedrus bidwillii Hook. f.	12.4ef	1.2
D. cupressinum	12.1f	0.6
Griselinia lucida Forst. f.	11.2fg	0.8
Litsea calicaris (A. Cunn.) Benth. et Hook. f. ex Kirk	10.0fg	0.6
Agathis australis Salisb.	9.9fg	0.7
Nothofagus solandri var. *solandri* (Hook. f.) Cheesem.	9.8fg	0.7
P. taxifolia	9.4g	0.9
Phyllocladus trichomanoides D. Don	9.4g	0.6
P. ferruginea	7.1h	0.5
B. tawa	4.7i	0.4

Values in a column followed by the same letter do not differ significantly (LSD test; α = 0.05).

3.5. Graveyard study

Sample half-lives estimated from the graveyard data are given in Table 6 for all the indigenous species tested in these trials, and show clear differences in durability between species. Decay rates from the graveyard study are based on time taken for sample stakes to break and cannot be directly compared with decay rates from the CWD study which are based on the rate of reduction in wood density, but it was hoped that species rankings from the two data sets would be compatible. However, species rankings were quite different between the two data sets and in general there was greater variation between species in the graveyard data than the CWD study. Of the five species well represented in both sets, in the graveyard data *N. fusca* was the most durable followed in order by *N. menziesii*, *D. cupressinum*, *P. taxifolia* and *B. tawa*. In the CWD study, dbh-adjusted decay rates for *P. taxifolia*, *N. fusca*, *N. menziesii*, *and B. tawa* were higher than for *D. cupressinum and P. ferruginea*. For these five species, there was a 7-fold difference between the fastest and slowest decay rates in the graveyard study, but only a 2-fold difference in the CWD study.

4. Discussion

Carbon sequestration by forest trees can be calculated from the gains in carbon due to tree recruitment and growth minus losses due to mortality and decay. Because decay rates for indigenous tree species were not known, a widely used stock change approach was adopted in New Zealand (Coomes et al., 2002), where carbon in CWD is estimated from its volume, which is then converted to carbon by classifying debris into decay classes, and applying decay class-specific density modifiers (Harmon and Sexton, 1996; Waddell, 2002). This approach is useful when undertaking an inventory of CWD carbon stocks, but has a number of shortcomings noted by Creed et al. (2004). CWD decay classes are subjectively assessed categorical data that are based on surface rather than internal log characteristics. These shortcomings may be expected to result in biased carbon stock change estimates where the objective is to estimate changes over annual or decadal timescales. Moreover, CWD density is logistically difficult to measure in large trees commonly found in old growth forest. The development and use of decay functions for predicting decay rates is clearly preferable.

Our study provides estimates of CWD decay rates based on estimates of the density of live trees and loss in mass due to insect tunneling and fungal decay approximately 20 years following windfall for seven important tree species growing in indigenous forests in the central North Island of New Zealand. The decay constants were fitted using a single-exponential decay model and apply to all fractions in the wood and bark of stems and branches (>10 cm), following Means et al. (1985). Decay constants were estimated using double sampling, where the survey data provided a precise estimate of the site mean outer wood basic density by species, and the recently fallen trees provided a moderately strongly correlation (r = 0.87) with whole stem and branch density. The ratio estimator gave precise estimates of the initial densities by tree species of the CWD prior to tree mortality, while the actual densities of CWD samples from dated storms were used to estimate the loss in density, which was clearly measurable after 20 years of decay. The effects of fragmentation of CWD on density were allowed for by using stem diameter data acquired previously for 16 trees of two species (*P. taxifolia* and *D. cupressinum*) that were measured four years after windfall (Hood et al., 1989), and for other species without prior stem diameter data by allowing for bark and sapwood loss, which could still be determined after 20 years, following Harmon and Sexton (1996).

Significant differences in decay constants among these seven tree species were found. When corrected to a standardized dbh, *P. taxifolia*, *N. fusca*, and *N. menziesii* decayed more slowly than *D. cupressinum*, *P. ferruginea*, and *D. dacrydioides*, while *B. tawa* had an intermediate decay rate. In addition to tree species, dbh also needs to be taken into account when estimating decay rates of individual trees, because the proportion of heartwood depends on tree size. Our observations suggest that previous sampling of CWD did not artificially influence decay rates, with cut heartwood faces, although moss covered, often sound for some sections. Furthermore, trees frequently break into sections naturally following windfall events.

Previous studies of CWD decomposition have been conducted in two beech forests in the South Island of New Zealand, one in mountain beech (*Nothofagus solandri* var. *cliffortioides*; Clinton et al., 1999; Allen et al., 2000; Buchanan et al., 2001), and the other in a red/silver beech forest (Stewart and Burrows, 1994). However, these studies did not include decay rates which are documented here for the first time for New Zealand indigenous species. Stewart and Burrows (1994) measured CWD densities, but reliable decay constants cannot be calculated because volume loss in the decayed logs was not assessed (Harmon and Sexton, 1996). However, their study does provide information on the longevity of the decay process in fallen trees. They assigned all fallen trees to four decay classes and estimated the mean age of each class by dating sudden growth rate increases in neighboring trees using increment cores. The CWD in the most advanced decay class representing the final stage of decomposition had an average age of 99 years. This result is consistent with our study; the exponential decay model with our estimate for λ of 0.023 predicts a mass loss after 100 years of 90%.

The decay rates reported in our study for New Zealand indigenous forest CWD are comparable to, or slightly lower, than those reported for temperate forests in other countries. For example, Chambers et al. (2000) collated rate constants from 20 studies in a number of countries including 16 North American temperate forest studies. They found decay rates to be highly correlated with temperature, and developed a model for predicting the decay constant as a function of MAT. At a MAT of 10 °C which corresponds to our study sites, their model predicts a decay constant of 0.038 giving a half-life of 18 years, and this can be compared to our estimate of 0.023 and half-life of 30 years.

For the specific sites we examined, CWD decay rates can be predicted given the tree species, dbh, and the fungal species present. Periodically measured permanent sample plots can provide the tree measurement and mortality data required to predict CWD stock changes over time using the models developed in this study. It remains to be seen how CWD decay rates for particular tree species vary in indigenous forests in other parts of New Zealand. However, site mean annual temperature is known to influence CWD decay rates with global data sets suggesting a Q_{10} of 2.53 (Mackensen et al., 2003), or 2.4 (Chambers et al., 2000), and it may be appropriate to apply this result in New Zealand for sites with temperatures significantly different from those used in this study. This could be achieved by multiplying the decay constant (e.g., predicted using Eq. (11)) by $Q_{10}^{(T-\bar{T})/10}$, where T is the MAT of the site and $\bar{T}$ is the MAT of the sites in this study (about 10 °C).

At least 20 decay fungi were isolated from the CWD trees at both locations, the most frequent species being *Ganoderma* cf. *applanatum*, *Armillaria* spp., and additionally on *Nothofagus* spp., *Cyclomyces tabacinus*. When analysed together, there was no significant relationship between yield of decay fungi and wood density 20 years after windfall. At this time, many stems, especially those of smaller diameter, were well colonized by decay fungi, giving rise to high isolation yields. However, when analysed separately, differences in wood density were found to be related to the fungal species present. In particular, *G.* cf. *applanatum* was associated with a more rapid loss in wood density and speedier decomposition than was found with other fungal species. Others have reported variation in decay rates among species and groups of decomposer fungi (Worrall et al., 1997; Edman et al., 2006; Buchanan et al., 2001). A model was fitted which potentially allows for refinement of the survey function by recognizing the prevalence of *G.* cf. *applanatum*, should this be advantageous. Colonisation by *G.* cf. *applanatum* is readily determined by the occurrence of a distinctive, hard, woody, perennial bracket commonly seen on CWD in indigenous forests. This study has shown that the presence or absence of such fungal fruit bodies is a reliable indicator of whether or not segments of CWD are colonized by *G.* cf. *applanatum* (Hood et al., 2004, 2008). Wood decomposed by this species is also fairly characteristic, appearing soft, light, and fibrous to spongy in texture.

It was intended that our research into decay rates of indigenous trees would be extended to include other species by incorporating data from a species durability rating system derived from in-ground graveyard tests. Using these data would have necessitated calibrating the system to account for carbon loss, once sufficient CWD decay data had been obtained. Unfortunately, it was found that the species rankings obtained from the two systems were not directly comparable, as also found in Australia (Mackensen et al., 2003), possibly because decomposition was effected under dissimilar conditions by different decay agents acting, for some species on contrasting wood components. Decay of partly buried wooden stakes tends to occur where moisture content is high near ground level as a result of colonization by non-basidiomycete soft rot fungi and bacteria which employ distinctive decay mechanisms not necessarily corresponding to those occurring in CWD in forests.

5. Conclusions

The mean exponential decay rate constant for CWD in indigenous forests located in the central North Island of New Zealand was 0.023 and the predicted time for CWD to lose 50% mass was 30 years. Decay constants were found to differ among species, with species decay times varying by up to ±50% from the overall mean. However, because of variability in decay rates between trees of the same species, and limited sample sizes for some species, these differences were mostly not statistically significant. The rate of decay was found to be influenced as much by the particular mix of fungal species present on the fallen tree as by the tree species, with one common decay fungus species (*G.* cf. *applanatum*) associated with a greatly increased decay rate. Decay rates were faster in smaller than in larger diameter CWD. Until more detailed data can be collected, it is recommended that decay constants derived in this study be assigned to tree species groups, and decay rates adjusted for tree diameter (Eq. (11)), and temperature effects when being applied to all indigenous tree species throughout New Zealand. Decay studies should be conducted elsewhere in New Zealand indigenous forests, and for a wider range of species, to determine the effects of differing environmental conditions on decay rates.

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